

PHYSICS INFORMED NEURAL NETWORK FOR DIFFUSION EQUATION



ORGANISATION : ML4SCII
PROJECT : GENIE

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GOAL

- To develop a fast calorimeter particle shower simulator [Fast high-fidelity emulators of GEANT4].
Fast Calorimeter Simulation Challenge 2022 [<https://calochallenge.github.io/homepage/>]

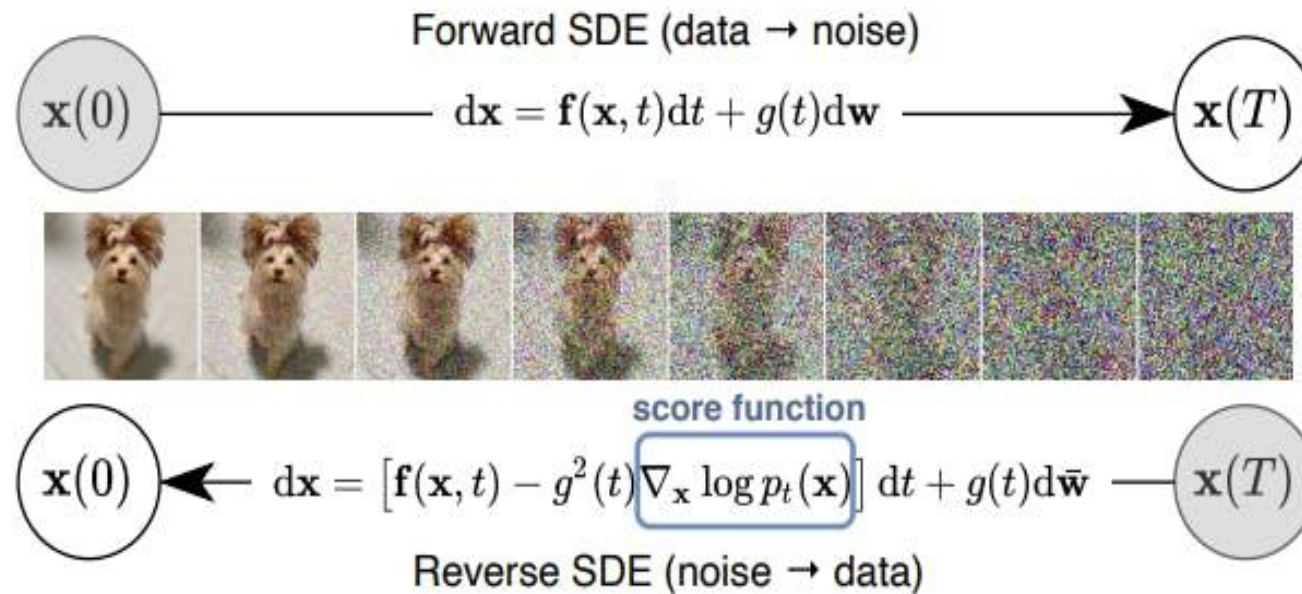
Sample from a distribution when we only have access to a finite number of samples from the distribution of Interest.



Generative models

DIFFUSION MODELS

Transform a distribution to gaussian distribution by gradually adding noise and learning to invert this process.



FORWARD STOCHASTIC DIFFERENTIAL EQUATION

$$dx_t = f(t)x_t dt + g(t)W_t, \quad x_0 \sim q_0(x_0)$$

With suitable choices , we can get the following conditional distribution

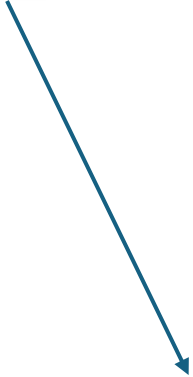
$$p(x_t|x_0) = \mathcal{N}(x_t; \alpha_t x_0, \sigma_t^2 \mathbb{I})$$

REVERSE-TIME ODE

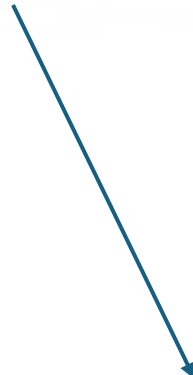
There exists an ODE that describes a deterministic process whose time-dependent distributions are exactly the same as the stochastic forward process at each time step.

$$dx_t = \left[f(t)x_t - \frac{1}{2}g^2(t)S(t, x_t) \right] dt, \quad S(t, x_t) \equiv \nabla_x \log p(x_t)$$

Our Approach :

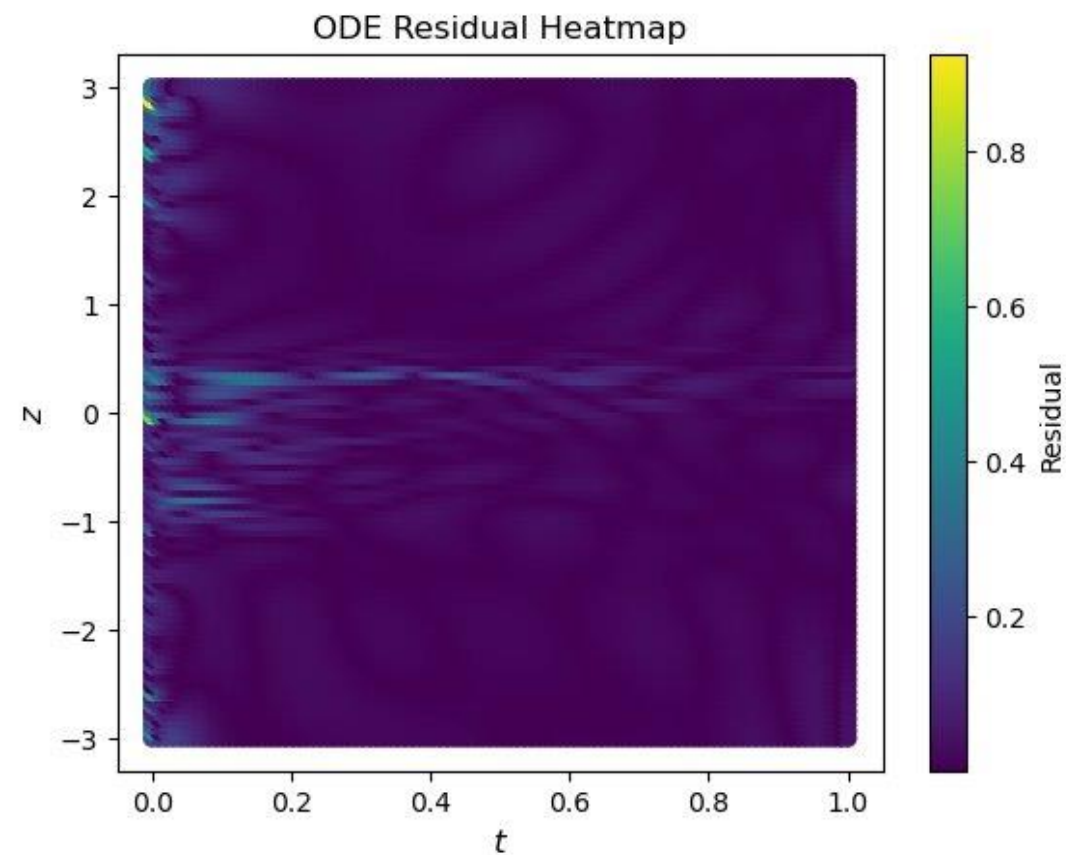
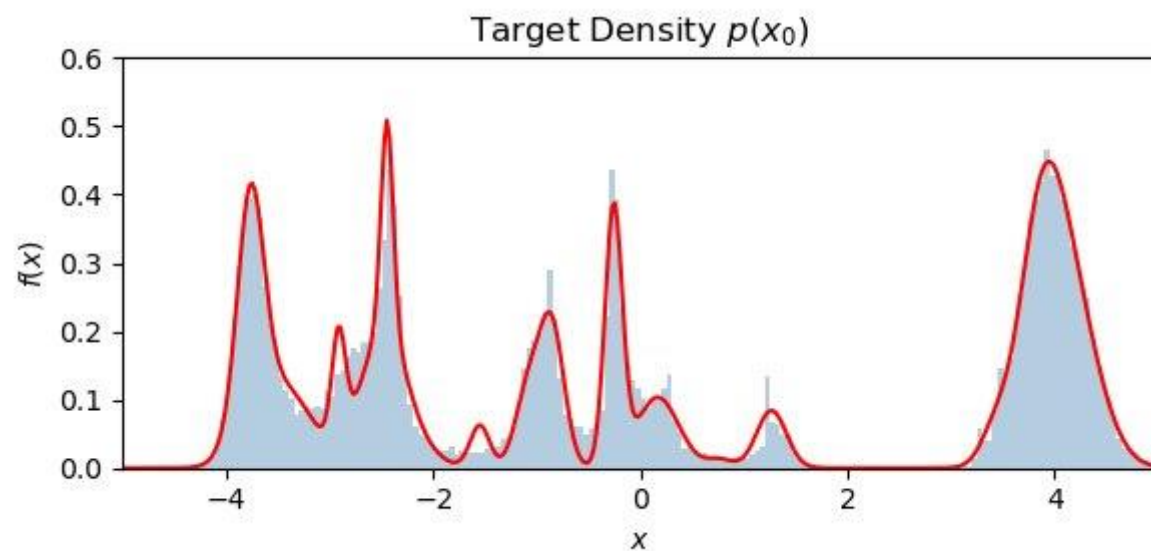


Solve the ODE
using PINN

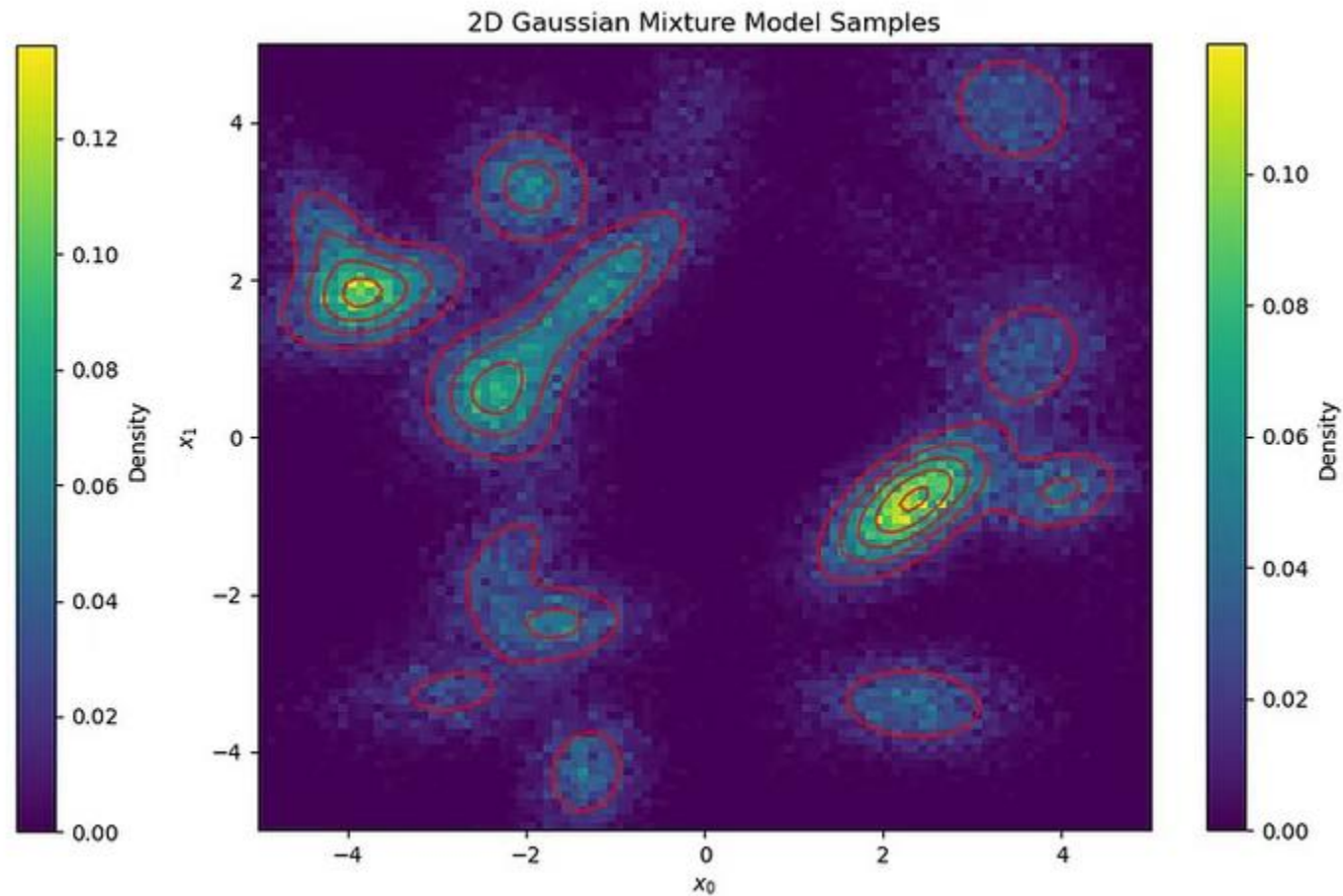
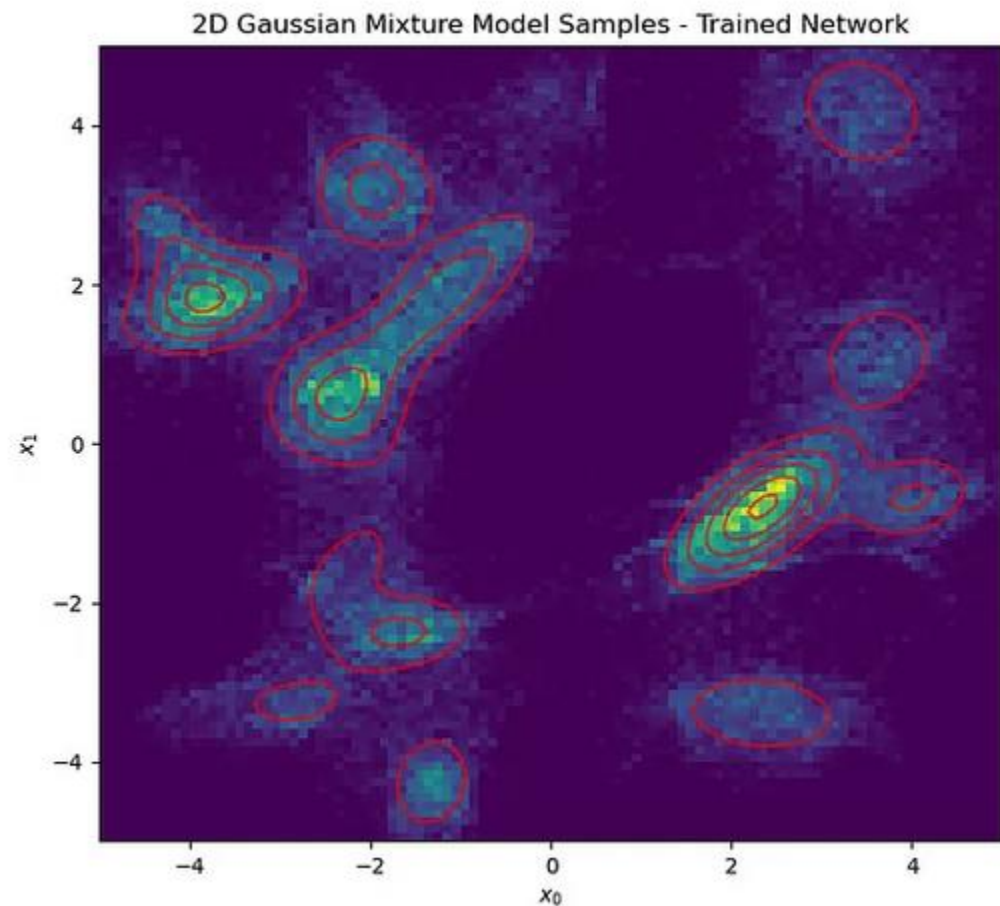


Approximate Score
function using Monte
Carlo Integral

PRELIMINARY RESULTS



PRELIMINARY RESULTS



PRELIMINARY RESULTS

